PowerWorld Scheduled Actions (Schedule) Data Summary

What “Scheduled Actions (Schedule)” is

Scheduled Actions is a **PowerWorld Simulator add-on** used to model and visualize **planned outages and other time-based actions** on the network. Its core purpose is to let you see the **combined effect of multiple scheduled changes**—for example, line outages, transformer outages, generator outages, switching actions, or other equipment state/value changes—over a time window.

In short, it is an **outage scheduling / planned switching analysis tool** rather than a contingency solver. It helps answer questions like:

- *What does the system look like when several planned outages overlap?*
- *What is in or out of service at a given date/time?*
- *How do sequential switching actions affect the case over a maintenance window?*

Main capabilities

1) Define time-based actions on model objects

You can schedule actions such as:

- **Open**
- **Close**
- **Modify object field values**

These actions are tied to times and can then be applied to the case so you can analyze the resulting network state.

So conceptually, a Scheduled Action is an instruction like:

- open a branch at time T1
- close it again at time T2
- take a generator out of service for a maintenance interval
- change a field value for a defined period

2) Analyze the

combined effects

of multiple planned outages

The tool is specifically meant to evaluate **overlapping or cumulative outage plans**. Rather than studying one outage at a time, you can build a time sequence of actions and then inspect the network as those actions accumulate over the selected time window.

3) Time navigation and animation

PowerWorld says users can:

- **manually scroll over a specified time period**, or
- **automatically animate** the scheduled actions being applied over that window.

This makes it useful for reviewing an outage plan chronologically and seeing which devices are out at each point in time.

4) Gantt-chart style overview

The Schedule add-on includes a **Gantt chart overview of Scheduled Action Groups**, which is useful for seeing outage overlaps and timing relationships across many planned actions.

Main data objects / concepts

The help page points to three related objects:

- **Scheduled Action**
- **Scheduled Action Group**
- **Scheduled Action Status**

A practical interpretation is:

Scheduled Action

A single action to be applied to some object at a particular time or over a defined interval.

Scheduled Action Group

A collection of related scheduled actions, likely used to represent one coordinated outage package, maintenance plan, switching sequence, or project window.

Scheduled Action Status

The current state / applicability of scheduled actions relative to the chosen time in the schedule analysis.

Typical use cases

The Schedule add-on is well suited for:

- **planned transmission outages**
- **generator maintenance scheduling**
- **switching plans / restoration sequences**
- **studying overlapping outages across a week or month**
- **visual outage coordination before operations or maintenance work**

Import / workflow support

PowerWorld indicates scheduled actions can be populated not only manually, but also via imports:

- the add-on supports **manual configuration**
- scheduled actions can be imported from **AUX**
- PowerWorld can import **CSV files exported from Control Room Operations Window (CROW)** outage coordination software.

For **CROW CSV import**, PowerWorld notes that device labeling conventions must match, so special labels may need to be created in the case to map CROW records to PowerWorld devices.

Notable fields / enhancements mentioned in PowerWorld help

PowerWorld's "What's New" notes mention a few Schedule-related items, including:

- a read-only field **"In Outage?"** on various objects to indicate whether the object is referenced by a current scheduled action
- ability to use a **Scheduled Action filter** on displays of Scheduled Action Groups
- a **Type** field on Scheduled Actions indicating outage type such as **transmission**, **generator**, or **generator adjustment**.

Bottom line

Scheduled Actions (Schedule) in PowerWorld is an **outage scheduling / planned-action timeline tool** for Simulator. It lets you:

- define **time-stamped network actions**
- group them into outage packages
- **step through or animate** the schedule over time
- visualize overlaps with a **Gantt chart**
- study the **compound network state** created by planned outages and switching operations.

If you want, I can do a second pass and turn this into a **more technical "data-model summary"** of the three objects **Scheduled Action / Scheduled Action Group / Scheduled Action Status**, including what fields they likely contain and how you'd map them into a planning database or CIM-style outage model.

Appendix Technical summary

1) Big picture: the Schedule add-on's object model

PowerWorld's **Scheduled Actions** feature is essentially a **time-based outage / switching model** layered on top of the base network case. The core object family is:

- **Scheduled Action Group**
- **Scheduled Action**
- **Scheduled Action Status**

The UI exposes these in the **Scheduled Actions Dialog**, with a time window (`Start Time`, `End Time`, `View Time`, `Resolution`) and a Gantt-style timeline for groups. Actions can be **applied**, **reverted**, animated over time, and filtered.

A useful mental model is:

- **Scheduled Action Group** = the **planned outage / work package / switching package**
- **Scheduled Action** = an **individual operation** against one device or one field
- **Scheduled Action Status** = the **status/state record** that says *when a group is active and how it should be treated operationally*

So if you were mapping this to a utility outage planning system:

- **Group** ≈ outage request / outage ticket / switching program / maintenance package
- **Action** ≈ one device-level step or one field modification
- **Status** ≈ lifecycle + effective outage window + approval/operational state

---

2) Object-by-object model

A. Scheduled Action Group = the outage container / schedule envelope

A **Scheduled Action Group** is the **top-level grouping object**. It collects related actions that belong to the same planned outage or switching event and defines the **time envelope** over which those actions are considered active.

Conceptual role

Use the Group to represent something like:

- “Take 230 kV line A–B out of service for maintenance from July 10 08:00 to July 10 17:00”
- “Transformer replacement outage package”
- “Substation switching sequence for breaker maintenance”
- “Generator outage plus associated switching and redispatch actions”

What the Group appears to own

From the WebHelp and UI behavior, the Group is the object that carries the **schedule window** and higher-level metadata for a planned outage package:

- **StartTime**
- **EndTime**
- **description / identifier fields**
- **status linkage**
- **time profile behavior**
- **the collection of child Scheduled Actions**

PowerWorld explicitly notes that the Group display now includes a **sub-grid of the Actions associated with the selected Group**, which strongly reinforces that the Group is the parent container.

Important Group behavior

1) Active over a time range

Groups are visualized in the Gantt chart as rows, with activity shown across time increments. The chart colors indicate whether a group is active/inactive during each time bucket.

2) Time profile

PowerWorld added **`TimeProfile`** to Scheduled Action Groups, with at least two modes:

- **Continuous** — active continuously between `StartTime` and `EndTime`
- **Daily** — active every day between the start day and end day, but only during the time-of-day window defined by StartTime/EndTime

That means a Group is not always a one-shot outage interval. It can model recurring daily windows such as:

- every day from 08:00–18:00 for a week
- day-shift maintenance outages
- repeated switching windows

Recommended interpretation for data modeling

If you were mapping PowerWorld to a planning database, **Scheduled Action Group** should be treated as the **master outage record** with fields roughly like:

Conceptual field	Meaning
`GroupID`	unique outage / schedule package identifier
`Name` / `Description`	human-readable outage title
`StartTime`, `EndTime`	planned effective interval
`TimeProfile`	continuous vs recurring-daily behavior
`StatusRef` / `CurrentStatus`	operational/lifecycle state
`Actions[]`	child actions belonging to the package

`Notes / Log / metadata`	optional operational notes

B. Scheduled Action = the atomic operation on a device or field

A **Scheduled Action** is the **individual action record** that actually changes the case. This is the object that targets a specific device or object field.

Conceptual role

An Action is one operation such as:

- **open** a line / breaker / branch
- **close** a line / breaker / branch
- **set** a field to a value
- **change** a field by an increment
- possibly apply switching isolation / energization logic around a target device

PowerWorld's own high-level description says actions may include **Open, Close, or manipulating object field values**. The CSV import page also confirms action mapping for **Open / Close / Set To / Change By** style actions.

---

3) What a Scheduled Action must contain conceptually

A Scheduled Action needs enough information to answer four questions:

(1)

What object is being changed?

The action targets a device or object in the case. From the CSV import help, device types commonly include:

- **Branches**
- **Generators**
- **Switched Shunts**
- **Loads**

and branch-like records can include things such as breakers or transformers depending on labeling/mapping.

So the action must carry some form of:

- **target object type**
- **target object identifier / key / label**

(2)

What operation should occur?

Typical action type is one of:

- **Open**
- **Close**
- **Set To**
- **Change By**

and possibly specialized breaker-oriented actions that later expand into lower-level open/close steps.

(3)

What value is involved, if any?

For `Set To` / `Change By`, a value field is required. PowerWorld's CSV import explicitly has a **Value Header** used for those action types.

(4)

When is the action active / should it be applied?

Operationally, the Action inherits timing from the active time evaluation of the Group / Status framework and from the current `View Time` / resolution window in the dialog.

---

4) Action fields you should expect in practice

Based on the help pages and "What's New" notes, a practical Scheduled Action record likely contains fields in these categories:

A) Parent linkage

- **Group reference / Group ID**
- possibly a row-level **outage ID** if imported from CSV

B) Target identification

- **object type** (`Branch`, `Gen`, `Load`, `SwitchedShunt`, etc.)
- **device label / key fields**
- possibly a **display label** or imported source label

C) Operation definition

- **Action Type**: `Open`, `Close`, `Set To`, `Change By`, etc.
- **Field** being modified, if this is a field-value action
- **Value** or **delta value**, if needed

D) Control / applicability fields

- **Allow Active** or equivalent enable flag
  PowerWorld mentions some automatically generated "extra" actions having `Allow Active = No`, meaning they are present informationally but not meant to execute by default.
- **Current Action Status**
  PowerWorld added a field showing the action's own **Current Status**, separate from the Group's current status.
- **Conflicts**
  The **Check Conflicts** function populates a **Conflicts** field when another action for the same device applies a different action.

E) Audit / provenance fields

- **Origin**
  PowerWorld documents `Origin` values such as:
  - `User` = directly user-created
  - `Added` = generated by breaker-identification logic
  - `Extra` = informational action also identified during breaker logic, usually not active by default
- **Log**
  PowerWorld added a read-only **Log** field to Scheduled Action objects for issues encountered when applying the action.

F) Classification fields

- **Type**
  PowerWorld notes a **Type** field used to classify outages, e.g.:
  - transmission
  - generator
  - generator adjustment

That is useful if you want to separate true switching outages from economic redispatch or generation maintenance actions.

---

5) Scheduled Action Status = the outage state / applicability layer

This is the least directly described object in the public WebHelp pages, but the behavior of the tool makes its role fairly clear.

A **Scheduled Action Status** appears to represent the **status record that determines how a Scheduled Action Group should be interpreted at a point in time**—for example whether the outage is active, inactive, planned, canceled, approved, etc., or how it should be included during time evaluation.

PowerWorld lists **Scheduled Action Status** as a first-class object alongside Group and Action, and the CSV import dialog says that if the CSV does not contain a field specifying a **Scheduled Outage Status**, a default can be assigned.

That tells us two things:

1. **Status is modeled explicitly**, not just implied by start/end timestamps.
2. The outage import workflow expects a status dimension from external outage systems.

Practical interpretation of Scheduled Action Status

I would treat this object as the **operational state / lifecycle state** attached to a Group, such as:

- Requested
- Approved
- Active
- Inactive
- Canceled
- Returned to service
- etc.

I can't safely claim those exact enum values without a page explicitly listing them, but the architecture strongly suggests this kind of role: a Group has a time window, and a Status object determines **whether and how that Group should be considered active in the case**.

Likely responsibilities of Status

A Scheduled Action Status probably contributes to:

- whether a group is considered **active at the current View Time**
- whether the outage is shown as active/inactive in displays
- whether imported outages from CSV are treated as eligible for application
- grouping outages by lifecycle state for operations planning

---

6) How application actually works in the simulator

The **Scheduled Actions Dialog** exposes the runtime behavior of this object model.

Time controls

The dialog has:

- **Start Time**
- **End Time**
- **View Time**
- **Resolution**
- **Animation controls**

These define the timeline window and the instant being evaluated.

Apply Actions

If **Apply Actions** is enabled, actions that are active at the current evaluation time are applied to the case. PowerWorld also stores/restores system state so the case can be rolled back when time changes or when the option is unchecked.

Use Normal Status

This tells Simulator to restore the **normal status** for scheduled actions that are not in the active range. In outage terms, this is how the case returns equipment to its pre-outage or nominal state when the outage window is not active.

Apply All Within Resolution

This is important operationally. Rather than applying only actions at one instant, PowerWorld can treat as active **all actions in the window from `ViewTime` to `ViewTime + one Resolution step`**.

So the evaluation engine is not strictly "single timestamp only"; it can work over a time bucket.

Evaluate Time Manually

This decouples moving the time slider from immediately applying actions. It is useful if you want to inspect or selectively evaluate specific points.

---

7) Breaker-aware behavior: the action layer is more than simple status flips

PowerWorld's Scheduled Actions model includes special support for **breaker operations**.

Identify Breakers

There is a tool to convert an **Open/Close Breakers** action into a set of individual open/close actions on actual devices. The generated actions are tagged with `Origin` such as `Added` or `Extra`.

That means the action model supports **high-level operational intents** ("open breakers for this device outage") that can be expanded into a detailed switching set.

Disconnect handling

The options include logic like:

- **Switch Disconnects to Normal Status in Open and Close Breakers Actions**

- additional handling in newer versions for disconnects/load-break disconnects in series when restoring or opening branches

So in practice, a Scheduled Action is not always just "set branch status = open." It can trigger topology-aware switching behavior around breakers and disconnects.

---

8) Import model: how external outage data maps in

The **CSV Import Dialog** is especially useful because it reveals the implicit schema PowerWorld expects.

Import has three conceptual mappings

A) Device Mapping

Maps each CSV row to a PowerWorld object via:

- **Type Header**
- device-type-specific values
- **Label Format / Import String Format**

This is how an outage system's equipment naming convention is translated into PowerWorld device keys.

B) Action Definition

Maps CSV action semantics to PowerWorld action types:

- `Open`
- `Close`
- `Set To`
- `Change By`

with an optional value column for the last two.

C) Outage Definition

Maps outage-level metadata:

- outage/group ID
- description
- **Scheduled Outage Status**
- **Start Time**
- **End Time**

PowerWorld explicitly says **Start Time and End Time are required** for scheduled outages.

Practical consequence

This import structure strongly supports the following relational interpretation:

`ScheduledActionGroup`

- `GroupID`
- `Description`
- `StartTime`
- `EndTime`
- `Status`
- `TimeProfile`

`ScheduledAction`

- `ActionID`
- `GroupID`
- `TargetObjectType`
- `TargetObjectKey`
- `ActionType`
- `FieldName`
- `Value`
- `Origin`
- `AllowActive`
- `Conflicts`
- `Log`

`ScheduledActionStatus`

- `StatusID`
- `StatusName`
- maybe flags for active/inactive applicability or lifecycle category

---

9) How I would map this into a utility planning / CIM-style model

If you want to normalize PowerWorld Scheduled Actions into your own outage-planning database, I would use a 3-level structure like this.

A. Outage / Schedule Package table

Equivalent to **Scheduled Action Group**

Suggested columns

- `outage_id`
- `name`
- `description`
- `status_code`
- `start_time`
- `end_time`
- `time_profile` (`continuous`, `daily`)
- `source_system` (`manual`, `AUX`, `CROW`, external CSV`)
- `notes`

B. Outage Step / Device Action table

Equivalent to **Scheduled Action**

Suggested columns

- `action_id`
- `outage_id`
- `sequence_no` if needed
- `target_type`
- `target_key`
- `action_type`
- `target_field`
- `set_value`

- `delta_value`
- `allow_active`
- `origin`
- `conflict_text`
- `log_text`
- `classification_type`

C. Status dimension / state table

Equivalent to **Scheduled Action Status**

Suggested columns

- `status_code`
- `status_name`
- `is_active_for_simulation`
- `lifecycle_stage`
- `description`

This gives you a clean bridge between:

- outage coordination systems like **CROW**
- PowerWorld Scheduled Actions
- a future enterprise outage or switching database

---

10) Operational distinction: Scheduled Actions vs Time-Step Schedules

One thing worth keeping separate: **Scheduled Actions** are not the same as the **Time Step Simulation "Schedules / Schedule Subscriptions"** feature.

PowerWorld also has a different scheduling mechanism where a **Schedule** is a list of `(DateTime, Value)` points used to drive object fields over time through **Schedule Subscriptions**. That mechanism is for things like changing load, generator output, or shunt status as a time-varying profile.

So the distinction is:

Scheduled Actions (Schedule add-on)

Best for:

- outage windows
- open/close switching actions
- planned maintenance outages
- coordinated equipment state changes over an outage timeline

Time Step Schedules / Schedule Subscriptions

Best for:

- load profiles
- generator dispatch profiles
- recurring MW/Mvar or status trajectories
- time-series simulation input

They are related in spirit, but **not the same data model**.

---

11) My distilled data-model view

If I compress PowerWorld Scheduled Actions into one sentence:

A Scheduled Action Group defines when an outage package is active, a Scheduled Action defines what device-level change to apply, and a Scheduled Action Status defines the outage/group's operational state used when evaluating that package over time.

That is the cleanest way to think about the model.

If you want, I can do one more pass and produce either of these:

1. **a concrete relational schema** for `ScheduledActionGroup / ScheduledAction / ScheduledActionStatus` with suggested SQL table definitions, or
2. **a CIM / EMS mapping** showing how PowerWorld Scheduled Actions correspond to outage records, switching steps, equipment status, and substation-level outage planning objects.